

Question 1

When configuring an IPv6 address on an interface, which of the following elements must be manually configured to ensure that the address is globally unique and can be routed across the internet?

- A. Prefix
- B. Global Routing Prefix
- C. Interface Identifier
- D. Subnet Identifier

Answer:

B. Global Routing Prefix

Explanation:

The **Global Routing Prefix** is assigned by an ISP and ensures that the IPv6 address is **globally unique and routable on the internet**. Without it, the address cannot be used for public communication. The other options are either automatically generated or used for internal structuring within a network.

Key Point:

Global Routing Prefix = Internet routability + global uniqueness

Question 2

A network administrator is configuring port security on a switch. Which of the following modes allows the switch to automatically remove now invalid secure MAC addresses from the CAM table after a specific period of time?

- A. Aging
- B. Protect
- C. Restrict
- D. Shutdown

Answer:

A. Aging

Explanation:

The **aging** feature in port security allows dynamically learned secure MAC addresses to be **removed automatically after a configured time period**. This helps keep the CAM table updated by clearing out inactive or outdated MAC addresses.

The other options are violation modes:

- **Protect** → Drops unauthorized traffic silently
- **Restrict** → Drops traffic and logs the violation
- **Shutdown** → Disables the port on violation

Key Point:

Aging = Automatically removes old secure MAC addresses over time

Question 3

In a network using OSPF (Open Shortest Path First), a router with multiple OSPF interfaces has varying load requirements and bandwidth configurations. Which of the following OSPF features allows the assignment of different cost metrics to different interfaces to influence route selection?

- A. OSPF Hello Timer
- B. OSPF Priority

- C. OSPF Area ID
- D. OSPF Cost

Answer:

D. OSPF Cost

Explanation:

OSPF Cost is the metric used to determine the best path for routing. Administrators can manually assign different cost values to interfaces to **influence route selection**, allowing traffic to prefer certain paths over others.

The other options serve different purposes:

- **Hello Timer** → Controls how often OSPF hello packets are sent
- **Priority** → Determines DR/BDR election
- **Area ID** → Identifies OSPF areas

Key Point:

OSPF Cost = Controls path selection by assigning interface metrics

Question 4

Which of the following Ethernet standards provides the longest distance for fiber optic communication while maintaining 10 Gbps data rate?

- A. 10GBASE-SR
- B. 10GBASE-LR
- C. 10GBASE-ER
- D. 10GBASE-T

Answer:

C. 10GBASE-ER

Explanation:

10GBASE-ER (Extended Reach) supports distances of up to **40 km** over single-mode fiber, making it the longest-range option among the given standards while maintaining a 10 Gbps data rate.

Other options:

- **10GBASE-SR** → Short range (~300–400 m, multimode fiber)
- **10GBASE-LR** → Long range (~10 km, single-mode fiber)
- **10GBASE-T** → Uses copper (not fiber), up to ~100 m

Key Point:

10GBASE-ER = Longest distance (~40 km) at 10 Gbps over fiber

Question 5

In a Cisco Catalyst switch, which command would you use to enable 802.1X authentication on a specific port?

- A. switchport mode access
- B. switchport port-security
- C. dot1x port-control auto
- D. authentication port-control auto

Answer:

D. authentication port-control auto

Explanation:

The command **authentication port-control auto** enables **802.1X authentication** on a switch port, allowing the port to transition between unauthorized and authorized states based on successful authentication.

Other options:

- **switchport mode access** → Sets the port as an access port
- **switchport port-security** → Enables port security (not 802.1X)
- **dot1x port-control auto** → Legacy/incorrect syntax for modern Cisco IOS

Key Point:

authentication port-control auto = Enables 802.1X on a switch port

Question 6

In an IPv6 network, which of the following address types is used to communicate with multiple devices, but packets are delivered to the nearest one (based on routing metrics)?

- A. Link-Local Unicast
- B. Multicast
- C. Anycast
- D. Global Unicast

Answer:

C. Anycast

Explanation:

Anycast addresses are assigned to multiple devices, but traffic sent to that address is delivered to the **nearest device** based on routing metrics. This is commonly used for services like DNS to improve performance and redundancy.

Other options:

- **Link-Local Unicast** → Used for communication within the same network segment
- **Multicast** → Sends packets to **all** members of a group
- **Global Unicast** → Unique address for a single interface

Key Point:

Anycast = One-to-nearest (best path delivery)

Question 7

Which of the following types of IPv6 addresses allows for only a single interface to be identified and reached by packets addressed to that address?

- A. Broadcast
- B. Anycast
- C. Unicast
- D. Multicast

Answer:

C. Unicast

Explanation:

A **Unicast** address identifies a **single interface**, and packets sent to it are delivered to exactly one destination.

Other options:

- **Broadcast** → Not used in IPv6
- **Anycast** → Delivered to the nearest device among many
- **Multicast** → Delivered to multiple devices in a group

Key Point:

Unicast = One-to-one communication

Question 8

A network administrator is troubleshooting an issue where IPv4 multicast packets are not properly forwarded across a particular network segment. Which of the following multicast routing protocol should be configured to allow proper forwarding of these packets?

- A. PIM
- B. OSPF
- C. EIGRP
- D. RIP

Answer:

A. PIM

Explanation:

PIM (Protocol Independent Multicast) is specifically designed to route **multicast traffic** across networks. It works independently of the underlying unicast routing protocol and enables proper forwarding of multicast packets.

Other options:

- **OSPF** → Unicast routing protocol

- **EIGRP** → Unicast routing protocol
- **RIP** → Unicast routing protocol

Key Point:

PIM = Required for multicast routing across networks

Question 9

What is the primary reason for implementing port security on a switch in a network environment?

- A. To increase the speed of data transfer within the local network.
- B. To provide a backup path in case of main path failure.
- C. To prevent unauthorized devices from connecting to the network by limiting the number of MAC addresses allowed on a port.
- D. To enable VLAN tagging on the switch port.

Answer:

C. To prevent unauthorized devices from connecting to the network by limiting the number of MAC addresses allowed on a port.

Explanation:

Port security is used to **restrict access to a switch port** by allowing only a specified number of MAC addresses. This helps prevent unauthorized devices from connecting and protects against attacks like MAC flooding.

Other options:

- **A** → Not related to security
- **B** → Related to redundancy (e.g., routing protocols)
- **D** → Related to VLAN configuration

Key Point:

Port Security = Limits MAC addresses to block unauthorized access

Question 10

In a Cisco network, you have configured a trunk link between two switches, and you want to ensure that only VLANs 10, 20, and 30 are allowed across the trunk. Which command would you use to achieve this?

- A. switchport trunk allowedvlan 10,20,30
- B. switchport trunk allowed vlan 10,20,30
- C. switchport trunk vlan allowed 10,20,30
- D. switchport trunk allowed vlan add 10,20,30

Answer:

B. switchport trunk allowed vlan 10,20,30

Explanation:

The correct Cisco IOS command to specify which VLANs are allowed on a trunk is:
switchport trunk allowed vlan 10,20,30

This command **overwrites the allowed VLAN list** with only VLANs 10, 20, and 30.

Other options:

- **A** → Incorrect syntax (missing space)
- **C** → Incorrect command order
- **D** → Adds VLANs to existing list (does not restrict to only these VLANs)

Key Point:

switchport trunk allowed vlan = Defines exactly which VLANs pass through the trunk

Question 11

You have configured DHCP snooping on your switches to improve network security. The DHCP snooping database needs to be stored persistently to prevent the switch from losing IP-to-MAC mappings upon reboot. Which command should be utilized to achieve this?

- A. `ip dhcp database bootflash:dhcp_db.txt`
- B. `ip dhcp snooping trust all`
- C. `ip dhcp snooping database flash:snoop_database.txt`
- D. `ip dhcp snooping binding persistent`

Answer:

C. `ip dhcp snooping database flash:snoop_database.txt`

Explanation:

The command **`ip dhcp snooping database flash:snoop_database.txt`** is used to **store DHCP snooping bindings persistently** in a file. This ensures that IP-to-MAC mappings are retained even after a switch reboot.

Other options:

- **A** → Incorrect command (not used for DHCP snooping database)
- **B** → Configures trusted ports, not database storage
- **D** → Invalid/incorrect command

Key Point:

DHCP Snooping Database = Saves IP-MAC bindings persistently across reboots

Question 12

Which of the following statements correctly describes the process and purpose of IEEE 802.1X in network access?

- A. IEEE 802.1X is a standard for VLAN tagging that helps in efficient use of LANs.
- B. IEEE 802.1X provides a framework for centralized user authentication via an authentication server before access is granted to the network.
- C. IEEE 802.1X is used to allocate IP addresses dynamically using a centralized DHCP server.
- D. IEEE 802.1X handles the establishment of VPN connections to ensure secure remote access.

Answer:

B. IEEE 802.1X provides a framework for centralized user authentication via an authentication server before access is granted to the network.

Explanation:

IEEE 802.1X is a **port-based network access control (NAC)** protocol that requires devices to **authenticate with a central authentication server (e.g., RADIUS)** before being granted access to the network.

Other options:

- **A** → Refers to VLAN tagging (802.1Q)
- **C** → Refers to DHCP
- **D** → Refers to VPN technologies

Key Point:

802.1X = Authenticate first, then get network access

Question 13

What is the purpose of the 'ip dhcp snooping trust' command when applied to a switch port, and what are the potential implications of not using it correctly?

- A. It ensures that only DHCP Discover and DHCP Offer messages are allowed on the port, blocking other DHCP message types to streamline DHCP operations.
- B. It allows the switch to trust DHCP messages from the connected device, enabling it to assign IP addresses automatically.
- C. It designates the port as trusted for DHCP messages, preventing DHCP spoofing attacks by ensuring that only legitimate DHCP servers can respond to requests.
- D. It allows the switch to bypass DHCP inspection on that port, improving network performance by reducing the overhead of monitoring DHCP traffic.

Answer:

C. It designates the port as trusted for DHCP messages, preventing DHCP spoofing attacks by ensuring that only legitimate DHCP servers can respond to requests.

Explanation:

The command **ip dhcp snooping trust** marks a port as **trusted**, allowing it to send DHCP server messages (like Offer and Ack). This is typically applied to ports connected to **legitimate DHCP servers or uplinks**.

If not configured correctly:

- Legitimate DHCP responses may be **blocked**
- Clients may **fail to obtain IP addresses**
- Network connectivity issues can occur

Other options:

- **A** → Incorrect description of DHCP filtering
- **B** → Misrepresents how DHCP works
- **D** → Incorrect; security is the main purpose, not performance

Key Point:

ip dhcp snooping trust = Allows only trusted ports to send DHCP server responses

Question 14

Which of the following methods uses a combination of symmetric and asymmetric encryption to ensure the confidentiality and authenticity of data in transit?

- A. Hash-Based Message Authentication Code (HMAC)
- B. SSL/TLS
- C. IPsec
- D. Secure Shell (SSH)

Answer:

B. SSL/TLS

Explanation:

SSL/TLS uses a combination of **asymmetric encryption** (for key exchange and authentication) and **symmetric encryption** (for fast data transmission). This hybrid approach ensures both **confidentiality and authenticity** of data in transit.

Other options:

- **HMAC** → Provides integrity and authentication, not encryption
- **IPsec** → Uses encryption but not typically described in this hybrid context for exams
- **SSH** → Uses encryption, but SSL/TLS is the standard answer for hybrid encryption model

Key Point:

SSL/TLS = Asymmetric (handshake) + Symmetric (data transfer)

Question 15

Which method should be used to securely manage network devices and ensure maximum encryption strength?

- A. SSH Version 1
- B. Telnet
- C. SSH Version 2
- D. HTTP

Answer:

C. SSH Version 2

Explanation:

SSH Version 2 (Secure Shell v2) provides **strong encryption, secure authentication, and integrity protection**, making it the preferred method for securely managing network devices.

Other options:

- **SSH Version 1** → Outdated and less secure
- **Telnet** → Sends data in plain text (not secure)
- **HTTP** → Not encrypted (use HTTPS instead)

Key Point:

SSH v2 = Secure remote access with strong encryption

Question 16

You have configured port security on a switch port. After a few hours, the port goes into the 'err-disabled' state despite no obvious security violations. Which of the following is the **LEAST** likely cause of this issue?

- A. The maximum number of secure MAC addresses has been exceeded.
- B. A sticky MAC address was manually removed from the configuration.
- C. MAC address flapping.
- D. A loopback interface has been created on a connected device.

Answer:

D. A loopback interface has been created on a connected device.

Explanation:

A **loopback interface** is a logical interface used for testing and internal communication within a device. It does **not generate traffic on a physical switch port**, so it would not trigger port security violations.

Other options are valid causes:

- **A** → Exceeding max MAC addresses triggers violation
- **B** → Removing sticky MAC can cause mismatch and violation
- **C** → MAC flapping (same MAC on different ports) can trigger security issues

Question 17

Which command would you use to enable an SNMP agent on a Cisco router and specify a read-only community string 'public' with access only from the subnet 192.168.10.0/24?

- A. `snmp-server community public ro 192.168.10.0 0.0.0.255`
- B. `snmp-server community public ro 192.168.10.0 0.0.255.255`

- C. snmp-server community public ro 192.168.10.0 0.0.0.255 access-list 10
- D. snmp-server community public ro 192.168.10.0 255.255.255.0

Answer:

A. snmp-server community public ro 192.168.10.0 0.0.0.255

Explanation:

The correct command uses a **wildcard mask** to define the subnet:

- **192.168.10.0 0.0.0.255** corresponds to **/24 subnet**

This command sets:

- **public** → community string
- **ro** → read-only access
- Restricts access to the **192.168.10.0/24 network**

Other options:

- **B** → Incorrect wildcard mask (/16)
- **C** → Incorrect syntax (ACL not properly defined here)
- **D** → Uses subnet mask instead of wildcard mask (invalid in this command)

Key Point:

SNMP uses wildcard mask → /24 = 0.0.0.255

Question 18

You have configured HSRP on a network and encounter a situation where both routers claim to be the active router at the same time, causing an IP address conflict. Which of the following HSRP settings can help you prevent such a situation by ensuring that a router always yields to the router with a higher priority?

- A. Authentication
- B. Interface Tracking
- C. Preemption Delay
- D. Priority Preemption

Answer:

D. Priority Preemption

Explanation:

Priority Preemption allows a router with a **higher priority** to take over as the active router automatically when it becomes available. This ensures proper role assignment and prevents multiple routers from acting as active at the same time.

Other options:

- **Authentication** → Secures HSRP communication
- **Interface Tracking** → Adjusts priority based on interface status
- **Preemption Delay** → Delays takeover, but doesn't enforce priority-based control

Key Point:

Preemption = Higher priority router takes control automatically

Question 19

In a secure network environment, which of the following security measures serves to isolate specific portions of your internal network to prevent unauthorized access from external hosts?

- A. VLAN (Virtual Local Area Network)
- B. IDS (Intrusion Detection System)
- C. VPN (Virtual Private Network)
- D. Network Firewall

Answer:

D. Network Firewall

Explanation:

A **Network Firewall** is used to **control and filter traffic between internal and external networks**, effectively isolating the internal network from unauthorized external access. It enforces security policies and blocks unwanted traffic.

Other options:

- **VLAN** → Segments internal network, not primarily for external protection
- **IDS** → Detects threats but does not block them
- **VPN** → Provides secure remote access, not isolation

Key Point:

Firewall = Protects internal network from external unauthorized access

Question 20

Which of the following mechanisms, while configuring an IPsec VPN, ensures data integrity and non-repudiation?

- A. AH with HMAC-SHA-2 and digital signatures
- B. ESP with 3DES encryption
- C. AH with MD5 hashing
- D. ESP with DES encryption

Answer:

A. AH with HMAC-SHA-2 and digital signatures

Explanation:

Authentication Header (AH) provides **data integrity and authentication**, and when combined with **HMAC-SHA-2 and digital signatures**, it ensures strong integrity and supports **non-repudiation**.

Other options:

- **ESP with 3DES** → Provides encryption (confidentiality), not non-repudiation
- **AH with MD5** → Weaker hashing, less secure
- **ESP with DES** → Outdated and insecure encryption

Key Point:

AH + Strong hashing + Digital signatures = Integrity + Non-repudiation

Question 21

In the context of Network Address Translation (NAT), what is the primary difference between Dynamic NAT and PAT (Port Address Translation)?

- A. Dynamic NAT allows multiple private IP addresses to share a single public IP address, while PAT assigns a unique public IP address to each private IP address.
- B. Dynamic NAT is also known as Overloading, while PAT is simply called Static NAT.
- C. Dynamic NAT maps a single private IP address to a single public IP address, while PAT maps multiple private IP addresses to a single public IP address using different ports.
- D. Dynamic NAT is used for translating IPv4 addresses, while PAT is used exclusively for translating IPv6 addresses.

Answer:

C. Dynamic NAT maps a single private IP address to a single public IP address, while PAT maps multiple private IP addresses to a single public IP address using different ports.

Explanation:

- **Dynamic NAT** → Maps **one private IP to one public IP** from a pool
- **PAT (Port Address Translation)** → Maps **multiple private IPs to a single public IP** using different port numbers (also called NAT overload)

Other options:

- **A** → Reversed definitions
- **B** → Incorrect terminology
- **D** → Both are used for IPv4

Key Point:

Dynamic NAT = 1:1 mapping

PAT = Many-to-1 using ports

Question 22

What is the primary purpose of the OSPF LSA Type-4 (ASBR Summary) in IP connectivity?

- A. It is used to advertise external routes redistributed into OSPF by an ASBR.
- B. It is used to suppress unnecessary routing updates.
- C. It is used to describe inter-area routes within OSPF.
- D. It is used to advertise the reachability of the ASBR.

Answer:

D. It is used to advertise the reachability of the ASBR.

Explanation:

LSA Type-4 (ASBR Summary LSA) is generated by ABRs to inform other areas **how to reach the ASBR (Autonomous System Boundary Router)**. This is necessary so routers can properly route traffic to external destinations advertised by the ASBR.

Other options:

- **A** → Refers to LSA Type-5 (External LSA)
- **B** → Not related to LSA Type-4
- **C** → Refers to LSA Type-3 (Summary LSA)

Key Point:

LSA Type-4 = Tells routers how to reach the ASBR

Question 23

You are configuring a Cisco router to provide Network Time Protocol (NTP) services to your network. The router must act as an NTP server while also synchronizing its time with an external NTP server at 192.168.1.10. Which set of commands correctly configures the router to achieve this and ensure secure NTP communication?

- A. ntp server 192.168.1.10 ntp peer 192.168.1.10 ntp master 3
- B. ntp server 192.168.1.10 ntp master 3 ntp trusted-key 1 ntp authenticate
- C. ntp server 192.168.1.10 ntp master 3 ntp trusted-key 1
- D. ntp server 192.168.1.10 ntp master 3 ntp trusted-key 1 ntp authentication-key 1 md5 NtpKey123 ntp authenticate

Answer:

D. ntp server 192.168.1.10 ntp master 3 ntp trusted-key 1 ntp authentication-key 1 md5 NtpKey123 ntp authenticate

Explanation:

This configuration:

- **ntp server 192.168.1.10** → Syncs with external NTP server
- **ntp master 3** → Allows router to act as an NTP server (stratum 3)
- **ntp authentication-key 1 md5 NtpKey123** → Defines authentication key
- **ntp trusted-key 1** → Marks the key as trusted
- **ntp authenticate** → Enables NTP authentication

Other options:

- **A** → Missing authentication
- **B** → Missing authentication key definition
- **C** → Incomplete secure configuration

Key Point:

Secure NTP = Authentication key + trusted key + authentication enabled

Question 24

Which protocol is used by IPv6 to discover other IPv6 devices on the same network and to configure addresses and parameters automatically?

- A. DHCPv6
- B. NDP
- C. ICMPv6
- D. ARP

Answer:

B. NDP

Explanation:

NDP (Neighbor Discovery Protocol) is used in IPv6 to:

- Discover other devices on the same network
- Determine link-layer addresses
- Perform address autoconfiguration (SLAAC)
- Identify routers and prefixes

It operates using **ICMPv6 messages**, but NDP is the actual protocol responsible for these functions.

Other options:

- **DHCPv6** → Used for address assignment, not discovery
- **ICMPv6** → Underlying protocol used by NDP
- **ARP** → Used in IPv4, not IPv6

Key Point:

NDP = IPv6 replacement for ARP + auto-configuration + neighbor discovery

Question 25

A network engineer is configuring SNMPv3 on a router. They have created a user with the username admin, set the group to manager, and specified the privacy and authentication protocols as AES and SHA respectively.

During testing, the SNMP manager is unable to poll the router. Which of the following could be the cause of the issue?

- A. The username 'admin' is not configured correctly.
- B. The access control list (ACL) is blocking SNMP traffic.
- C. The SNMP manager does not support SNMPv3.
- D. The time on the router and SNMP manager is not synchronized.

Answer:

B. The access control list (ACL) is blocking SNMP traffic.

Explanation:

A common issue when SNMP polling fails is that an **ACL is blocking SNMP traffic (UDP port 161)**. Even if SNMPv3 is configured correctly with authentication and encryption, traffic must still be **permitted through access control policies**.

Other options:

- **A** → Possible but less likely if configuration was successful
- **C** → Possible, but typically verified beforehand
- **D** → Time sync is not required for SNMPv3 operation

Key Point:

SNMP not working? → Always check ACLs (UDP 161) first

Question 26

In a network with an 802.1X authentication system, which of the following components is responsible for verifying the credentials of devices trying to connect to the network and granting or denying access based on the policies configured?

- A. Radius Server
- B. ACL
- C. Supplicant
- D. Authenticator

Answer:

A. Radius Server

Explanation:

In **802.1X** architecture:

- **Supplicant** → The client device requesting access
- **Authenticator** → The switch/AP controlling access
- **RADIUS Server** → The **authentication server** that verifies credentials and decides whether access is granted or denied

Other options:

- **ACL** → Used for traffic filtering, not authentication
- **Supplicant** → Requests access, does not verify credentials
- **Authenticator** → Passes authentication requests, does not verify

Key Point:

RADIUS Server = Verifies credentials and enforces access decisions

Question 27

Which of the following methods provides the most secure means to prevent a Man-In-The-Middle (MITM) attack during an SSL/TLS handshake?

- A. Employing Perfect Forward Secrecy (PFS).
- B. Utilizing a public Certificate Authority (CA) for certificate issuance.
- C. Implementing Pre-Shared Keys (PSK) for SSL/TLS sessions.
- D. Using self-signed certificates.

Answer:

B. Utilizing a public Certificate Authority (CA) for certificate issuance.

Explanation:

A **public Certificate Authority (CA)** ensures that certificates are **trusted and verified**, allowing clients to confirm the identity of the server. This prevents attackers from impersonating a legitimate server, which is the core of **MITM attacks**.

Other options:

- **PFS** → Protects past sessions, not identity verification
- **PSK** → Secure but not scalable and less commonly used
- **Self-signed certificates** → Not trusted by default, vulnerable to MITM

Key Point:

CA-signed certificates = Prevent MITM by verifying identity

Question 28

In a Layer 3 network, which of the following protocols is used to discover and maintain subnet router advertisements, and what role does it serve?

- A. DHCP, to dynamically assign IP addresses to devices in the network.
- B. BGP, to exchange routing information between different Autonomous Systems.
- C. OSPF, to perform route computation based on the OSPF algorithm.
- D. NDP, to handle neighbor discovery and solicitation in IPv6 networks.

Answer:

D. NDP, to handle neighbor discovery and solicitation in IPv6 networks.

Explanation:

NDP (Neighbor Discovery Protocol) is responsible for:

- Router advertisements (RA)

- Router discovery
- Neighbor solicitation and advertisement
- IPv6 address configuration (SLAAC)

It plays a key role in **IPv6 Layer 3 operations** by helping devices discover routers and network parameters.

Other options:

- **DHCP** → Assigns IP addresses
- **BGP** → Inter-AS routing
- **OSPF** → Interior routing protocol

Key Point:

NDP = Router advertisements + IPv6 neighbor discovery

Question 29

Which of the following technologies provides network access control, ensuring that only properly authenticated and compliant devices access the network, and helps to prevent threats by isolating non-compliant devices?

- A. MAC address filtering
- B. 802.1X
- C. NAC
- D. VPN

Answer:

C. NAC

Explanation:

NAC (Network Access Control) enforces **security policies** by ensuring that only **authenticated and compliant devices** can access the network. It can also **isolate or remediate non-compliant devices**, providing a higher level of control and protection.

Other options:

- **MAC address filtering** → Weak security, easily spoofed
- **802.1X** → Provides authentication, but NAC includes compliance checks and enforcement
- **VPN** → Provides secure remote access, not device compliance control

Key Point:

NAC = Authentication + Compliance check + Isolation of non-compliant devices

Question 30

You are configuring Network Address Translation (NAT) on a Cisco router. Which configuration ensures that internal IP addresses are translated into a single public IP address while retaining unique sessions for clients accessing external networks?

- A. Static NAT
- B. Dynamic NAT
- C. Port Address Translation (PAT)
- D. Dynamic PAT

Answer:

C. Port Address Translation (PAT)

Explanation:

PAT (Port Address Translation) allows **multiple internal IP addresses to share a single public IP address** by using **different port numbers** to maintain unique sessions for each client.

Other options:

- **Static NAT** → One-to-one mapping
- **Dynamic NAT** → Uses a pool of public IPs (one-to-one per session)
- **Dynamic PAT** → Not a standard term (PAT already implies dynamic behavior)

Key Point:

PAT = Many users → One public IP (using ports)

Question 31

What is the primary purpose of a next-generation firewall (NGFW) in a network security architecture?

- A. To serve as a load balancer and manage traffic distribution across servers.
- B. To perform network address translation (NAT) and hide internal network structure.
- C. To provide basic packet filtering based on IP addresses and port numbers.
- D. To integrate application awareness, in-line deep packet inspection (DPI), and threat intelligence.

Answer:

D. To integrate application awareness, in-line deep packet inspection (DPI), and threat intelligence.

Explanation:

A **Next-Generation Firewall (NGFW)** goes beyond traditional firewalls by providing:

- **Application awareness and control**
- **Deep Packet Inspection (DPI)**
- **Integrated threat intelligence and intrusion prevention**

This enables more advanced and intelligent security decisions.

Other options:

- **A** → Load balancer function
- **B** → NAT function (basic firewall feature)
- **C** → Traditional firewall capability

Key Point:

NGFW = DPI + Application awareness + Advanced threat protection

Question 32

Suppose you have a network topology with multiple routers connected through serial links and running OSPF as the routing protocol. All routers are correctly configured with the OSPF protocol. When you examine the OSPF neighbor table on one of the routers, you observe that one of its neighbors is stuck in the ‘2-way’ state and does not transition to the ‘Full’ state. What is the most likely cause of this issue?

- A. One of the serial links is experiencing intermittent physical layer failures.
- B. The OSPF router ID of the neighbor is the same as another router in the topology.
- C. There is a problem with the MTU size settings on the interfaces between the two routers.
- D. The routers are on a broadcast or NBMA network, and a DR/BDR election has occurred where the stuck router is not the DR or BDR.

Answer:

D. The routers are on a broadcast or NBMA network, and a DR/BDR election has occurred where the stuck router is not the DR or BDR.

Explanation:

In OSPF, neighbors on **broadcast or NBMA networks** do not always form a **Full** adjacency with every other router. Instead, routers usually form full adjacencies only with the **Designated Router (DR)** and **Backup Designated Router (BDR)**.

If a router remains in the **2-way** state, it often means:

- Bidirectional communication is working
- The routers see each other
- But they are **not required to become fully adjacent**

Other options:

- **A** → Physical issues usually cause adjacency loss or instability, not a normal 2-way state
- **B** → Duplicate router IDs cause OSPF problems, but not typically a normal 2-way condition
- **C** → MTU mismatch usually causes neighbors to get stuck in **Exstart** or **Exchange**, not 2-way

Key Point:

OSPF 2-way state is normal on broadcast/NBMA networks for routers that are not the DR or BDR.

Question 33

Which of the following IPv6 multicast addresses is used as the all-router multicast address?

- A. FF02::5
- B. FF02::6
- C. FF02::2
- D. FF02::1

Answer:

C. FF02::2

Explanation:

The IPv6 multicast address **FF02::2** is the **all-routers multicast address**, meaning all IPv6-enabled routers on the local link listen to this address.

Other options:

- **FF02::1** → All-nodes multicast address
- **FF02::5** → OSPFv3 AllSPFRouters
- **FF02::6** → OSPFv3 AllDRouters

Key Point:

FF02::2 = All routers on the local network

Question 34

Which of the following security protocols provides encryption at the Network Layer and is commonly used for Virtual Private Networks (VPNs)?

- A. SSH
- B. HTTPS
- C. SSL/TLS
- D. IPsec

Answer:

D. IPsec

Explanation:

IPsec (Internet Protocol Security) operates at the **Network Layer (Layer 3)** and provides:

- **Encryption (confidentiality)**
- **Integrity**
- **Authentication**

It is widely used to establish **VPN tunnels** between networks or devices.

Other options:

- **SSH** → Application layer (remote access)
- **HTTPS** → Application layer (web security)
- **SSL/TLS** → Transport layer security

Key Point:

IPsec = Network Layer encryption for VPNs

Question 35

In an 802.1X architecture for providing network access control, which of the following statements is true if a supplicant fails to respond to an EAP-Request/Identity frame sent by the authenticator?

- A. The authenticator automatically grants network access to the supplicant.
- B. The authenticator sends an EAP-Failure frame to the supplicant and terminates the connection attempt.
- C. The supplicant is added to a quarantine VLAN for limited network access until it responds to subsequent EAP-Request frames.
- D. The authenticator re-sends the EAP-Request/Identity frame for a pre-configured number of attempts.

Answer:

D. The authenticator re-sends the EAP-Request/Identity frame for a pre-configured number of attempts.

Explanation:

In **802.1X**, if the supplicant does not respond to an **EAP-Request/Identity**, the authenticator will **retry sending the request multiple times** based on configured timers before eventually failing the authentication.

Other options:

- **A** → Access is never granted without authentication
- **B** → Failure is sent only after retries are exhausted
- **C** → Quarantine VLAN is not the default behavior

Key Point:

No response → Authenticator retries EAP request before failing

Question 36

In the context of securing network traffic, which of the following mechanisms is the most effective way to ensure both origin authentication and data integrity in message transmissions?

- A. Symmetric Encryption
- B. Public Key Infrastructure (PKI)
- C. Transport Layer Security (TLS)
- D. HMAC (Hash-based Message Authentication Code)

Answer:

D. HMAC (Hash-based Message Authentication Code)

Explanation:

HMAC provides both:

- **Data integrity** → Ensures the message has not been altered
- **Origin authentication** → Confirms the sender possesses the shared secret key

Other options:

- **Symmetric Encryption** → Provides confidentiality, not integrity/authentication
- **PKI** → Manages keys and certificates, not direct message integrity
- **TLS** → Uses multiple mechanisms, but HMAC specifically ensures integrity and authentication

Key Point:

HMAC = Integrity + Authentication using a shared secret

Question 37

In a network security scenario, which of the following security mechanisms are primarily used to protect against unauthorized access and ensure confidentiality within a VLAN environment?

- A. Access Control Lists (ACLs)
- B. DHCP Snooping
- C. 802.1X Authentication
- D. Port Security

Answer:

A. Access Control Lists (ACLs)

Explanation:

ACLs (Access Control Lists) are used to **control traffic flow between VLANs and network segments**, helping prevent unauthorized access and enforce security policies. They can restrict which devices or users can communicate, thereby enhancing confidentiality.

Other options:

- **DHCP Snooping** → Protects against rogue DHCP servers
- **802.1X** → Provides authentication before access
- **Port Security** → Limits MAC addresses on a port

Key Point:

ACLs = Control access between VLANs and protect network resources

Question 38

In a HSRP (Hot Standby Router Protocol) network configuration, which of the following IP addresses would a host communicate with if it is configured to use the virtual IP address, while the primary router fails and a secondary router takes over?

- A. The virtual IP address configured for HSRP
- B. The MAC address of the secondary router
- C. The IP address of the failed primary router
- D. The IP address of the secondary router

Answer:

A. The virtual IP address configured for HSRP

Explanation:

In **HSRP**, hosts always communicate using the **virtual IP address**, which remains the same regardless of which router is active. When the primary router fails, the standby router takes over **without changing the IP address seen by hosts**.

Other options:

- **B** → Hosts use IP, not MAC directly
- **C** → Primary router is down
- **D** → Hosts do not use physical router IPs

Key Point:

HSRP = One virtual IP, multiple routers (seamless failover)

Question 39

In a switched LAN that uses 802.1X for port authentication, what role does the Authenticator play in the Network Access Control process?

- A. The Authenticator is the client device that requests access to the network.
- B. The Authenticator is the network device that enforces authentication before allowing network access.
- C. The Authenticator provides encryption keys to secure the communication between the client device and the network.
- D. The Authenticator is the server that validates the credentials of the client device.

Answer:

B. The Authenticator is the network device that enforces authentication before allowing network access.

Explanation:

In **802.1X architecture**:

- **Supplicant** → Client device requesting access
- **Authenticator** → Switch or access point that **controls access to the network**
- **Authentication Server (RADIUS)** → Validates credentials

The **Authenticator** acts as a gatekeeper, allowing or denying access based on authentication results.

Other options:

- **A** → Describes the supplicant
- **C** → Not the primary role of authenticator
- **D** → Describes the RADIUS server

Key Point:

Authenticator = Controls access (acts as gatekeeper)

Question 40

What is the primary function of the Service Level Agreement (SLA) monitor in a Cisco network environment?

- A. To measure network performance parameters such as latency, jitter, and packet loss
- B. To dynamically allocate bandwidth based on the real-time traffic load
- C. To establish encrypted connections between different network segments
- D. To provide redundancy and failover mechanisms for critical network services

Answer:

A. To measure network performance parameters such as latency, jitter, and packet loss

Explanation:

Cisco **IP SLA** is used to **monitor and measure network performance**, including:

- **Latency (delay)**
- **Jitter (variation in delay)**
- **Packet loss**
- **Availability of services**

This helps administrators **troubleshoot and ensure network performance meets requirements**.

Other options:

- **B** → QoS function
- **C** → VPN/IPsec function
- **D** → Redundancy protocols (e.g., HSRP, VRRP)

Key Point:

IP SLA = Monitors network performance and availability

Question 41

What type of attack primarily aims to exploit vulnerabilities in the fragmentation implementation of the IP protocol, potentially causing a system to crash?

- A. Teardrop Attack
- B. Smurf Attack
- C. Land Attack
- D. Ping of Death

Answer:

A. Teardrop Attack

Explanation:

A **Teardrop Attack** exploits weaknesses in how systems handle **IP packet fragmentation**. It sends malformed or overlapping fragments, causing the target system to **crash or become unstable** when trying to reassemble them.

Other options:

- **Smurf Attack** → ICMP amplification attack
- **Land Attack** → Sends packets with same source and destination IP
- **Ping of Death** → Sends oversized ICMP packets

Key Point:

Teardrop = Fragmentation attack that crashes systems

Question 42

Which command would you use to troubleshoot intermittent connectivity issues in a Cisco router and debug IP packets, while limiting the output to specific types of traffic?

- A. debug ip packet
- B. debug ip packet 100
- C. debug ip traffic
- D. debug ip packet detail

Answer:

B. debug ip packet 100

Explanation:

The command **debug ip packet 100** uses an **access control list (ACL 100)** to filter and limit the debug output to **specific traffic**, making troubleshooting more efficient and preventing excessive logging.

Other options:

- **debug ip packet** → Debugs all packets (can overwhelm CPU)
- **debug ip traffic** → Not a valid command
- **debug ip packet detail** → More verbose but not filtered

Key Point:

debug ip packet + ACL = Targeted troubleshooting without overload

Question 43

In a scenario where you need to extend VLANs across multiple switches, which advanced spanning-tree protocol would be most effective in avoiding bridge loops while also optimizing for rapid convergence, and why?

- A. PVST+ (Per-VLAN Spanning Tree Plus)
- B. MSTP (Multiple Spanning Tree Protocol)
- C. STP (Spanning Tree Protocol)
- D. RSTP (Rapid Spanning Tree Protocol)

Answer:

B. MSTP (Multiple Spanning Tree Protocol)

Explanation:

MSTP allows multiple VLANs to be mapped to a **single spanning-tree instance**, reducing overhead while still providing **rapid convergence** similar to RSTP. This makes it ideal for **large, multi-VLAN environments** where efficiency and scalability are important.

Other options:

- **PVST+** → One instance per VLAN (less scalable)
- **STP** → Slow convergence
- **RSTP** → Fast convergence but not optimized for multiple VLAN scaling like MSTP

Key Point:

MSTP = Scalable + Fast convergence for multi-VLAN networks

Question 44

Which of the following IEEE standards is used for frame tagging in VLANs?

- A. 802.11
- B. 802.1D
- C. 802.3
- D. 802.1Q

Answer:

D. 802.1Q

Explanation:

IEEE 802.1Q is the standard used for **VLAN tagging**, which inserts a tag into Ethernet frames to identify the VLAN they belong to when traversing trunk links.

Other options:

- **802.11** → Wireless networking (Wi-Fi)
- **802.1D** → Spanning Tree Protocol (STP)
- **802.3** → Ethernet standard

Key Point:

802.1Q = VLAN tagging standard

Question 45

Which statement accurately describes the behavior of the ‘ip nat inside source static’ command in a NAT configuration on a Cisco router?

- A. It only allows for outbound traffic from the inside network to the outside network, without permitting return traffic.
- B. It requires the specification of access control lists (ACLs) to function.
- C. It maps an inside local IP address to an inside global IP address, and automatically allows for bidirectional traffic.
- D. It maps multiple inside local IP addresses to a single inside global IP address, effectively providing PAT.

Answer:

C. It maps an inside local IP address to an inside global IP address, and automatically allows for bidirectional traffic.

Explanation:

The command **ip nat inside source static** creates a **one-to-one mapping** between:

- **Inside local IP** (private address)
- **Inside global IP** (public address)

This allows **both inbound and outbound traffic**, making it useful for hosting internal services accessible from outside.

Other options:

- **A** → Incorrect (static NAT allows return traffic)
- **B** → ACLs are not required for static NAT
- **D** → Describes PAT, not static NAT

Key Point:

Static NAT = 1:1 mapping with bidirectional communication

Question 46

When configuring Network Address Translation (NAT) on a router, which command would be used to establish a static translation between an inside local address of 192.168.10.5 and an inside global address of 203.0.113.5?

- A. `ip nat outside source static 192.168.10.5 203.0.113.5`
- B. `ip nat inside source list 1 interface FastEthernet0/1 overload`
- C. `ip nat inside source static 192.168.10.5 203.0.113.5`
- D. `ip nat pool NAT_POOL 192.168.10.5 203.0.113.5 netmask 255.255.255.0`

Answer:

C. `ip nat inside source static 192.168.10.5 203.0.113.5`

Explanation:

The command **`ip nat inside source static`** is used to create a **static one-to-one mapping** between:

- **Inside local IP** → 192.168.10.5
- **Inside global IP** → 203.0.113.5

This allows consistent inbound and outbound communication.

Other options:

- **A** → Used for outside NAT (different purpose)
- **B** → Used for PAT (overload), not static NAT
- **D** → Defines a NAT pool, not a static mapping

Key Point:

ip nat inside source static = Direct 1:1 IP mapping

Question 47

When configuring a VLAN trunk link that tags traffic with IEEE 802.1Q, which one of the following actions will ensure proper communication between switches if the native VLAN is mismatched?

- A. Disable the trunking on one side of the link
- B. Use VLAN Trunking Protocol (VTP) to synchronize VLAN configurations
- C. Enable Dynamic Trunking Protocol (DTP) on both switches
- D. Reconfigure the native VLAN to match on both ends of the trunk link

Answer:

D. Reconfigure the native VLAN to match on both ends of the trunk link

Explanation:

In **802.1Q trunking**, the **native VLAN is not tagged**. If there is a mismatch between switches, it can cause:

- Traffic misinterpretation
- VLAN leakage
- Connectivity issues

To fix this, both ends of the trunk must have the **same native VLAN configured**.

Other options:

- **A** → Disables trunking (not a solution)
- **B** → VTP does not fix native VLAN mismatch
- **C** → DTP negotiates trunking but doesn't resolve mismatch

Key Point:

Native VLAN must match on both sides of a trunk for proper communication

Question 48

Which of the following IPv6 multicast addresses is used for all routers on a local segment?

- A. FF02::1:FFAA:AA01
- B. FF02::2
- C. FF02::1
- D. FF02::5

Answer:

B. FF02::2

Explanation:

The IPv6 multicast address **FF02::2** is the **all-routers multicast address**, meaning all IPv6 routers on the local link listen to this address.

Other options:

- **FF02::1:FFAA:AA01** → Solicited-node multicast address
- **FF02::1** → All-nodes multicast address

- **FF02::5** → OSPFv3 AllSPFRouters

Key Point:

FF02::2 = All routers on the local network

Question 49

In the context of IPv6, which of the following statements is TRUE regarding the use of Neighbor Discovery Protocol (NDP)?

- A. NDP replaces both ARP and ICMP in IPv6 networks, handling all address resolution and error messaging.
- B. NDP uses Router Solicitations and Router Advertisements to discover routers and determine parameters like the subnet prefix and the MTU.
- C. NDP includes multicast listening discovery to determine which nodes on the link are still reachable.
- D. NDP is used for address autoconfiguration, but it requires the use of DHCPv6 for address stateful configuration.

Answer:

B. NDP uses Router Solicitations and Router Advertisements to discover routers and determine parameters like the subnet prefix and the MTU.

Explanation:

NDP (Neighbor Discovery Protocol) uses:

- **Router Solicitation (RS)** → Sent by hosts to discover routers
- **Router Advertisement (RA)** → Sent by routers to provide network info like prefix, MTU, and gateway

Other options:

- **A** → NDP replaces ARP, but ICMPv6 is still used (NDP is built on ICMPv6)
- **C** → Refers to MLD (Multicast Listener Discovery), not NDP
- **D** → NDP supports SLAAC without requiring DHCPv6

Key Point:

NDP = RS + RA for router discovery and auto-configuration

Question 50

Which of the following statements about IPv6 and IPv4 addressing is true?

- A. IPv6 headers are more complex than IPv4 headers due to the increased number of fields.
- B. IPv6 has built-in security features through IPsec, whereas IPv4 security features are optional.
- C. IPv6 uses 48-bit addresses, whereas IPv4 uses 32-bit addresses.
- D. IPv4 supports more addressable devices compared to IPv6 due to its NAT capabilities.

Answer:

B. IPv6 has built-in security features through IPsec, whereas IPv4 security features are optional.

Explanation:

IPv6 was designed with IPsec support built-in, making security a core part of the protocol. In contrast, **IPv4 can use IPsec**, but it is optional and not inherently integrated.

Other options:

- **A** → IPv6 headers are actually simpler than IPv4
- **C** → IPv6 uses **128-bit addresses**, not 48-bit
- **D** → IPv6 supports vastly more devices than IPv4

Key Point:

IPv6 = Built-in IPsec + larger address space (128-bit)